

Advaya Verma

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EDUCATION

Vellore Institute of Technology, Andhra Pradesh — B.Tech, Computer Science and Engineering | 2023 – 2027
CGPA: 8.1

Relevant Coursework: Data Structures & Algorithms, Operating Systems, DBMS, OOP, Machine Learning, Artificial Intelligence, Software Engineering & UML, Theory of Computation, Compiler Design, Design & Analysis of Algorithms

EXPERIENCE

AI Engineering Intern — IDM Valley | Apr 2026 – Jun 2026

- Applied semantic search and information-retrieval concepts from prior project work to develop and experiment with Retrieval-Augmented Generation (RAG) workflows for LLM-based applications.
- Worked with text embeddings, document retrieval, and vector databases to build retrieval pipelines for supplying relevant context to language models.
- Experimented with prompt engineering and context integration to improve the quality and relevance of LLM-generated responses.
- Used Python for data preprocessing, retrieval workflow development, experimentation, and integration of AI components with application backends.
- Created technical documentation covering RAG workflows, implementation approaches, configuration, and usage guidelines to support development and knowledge sharing.

PROJECTS

Semantic Search System | Python, FastAPI, FAISS, Docker, NLP, Scikit-learn | [GitHub](#)

- Built a semantic search pipeline using Sentence Transformers (MiniLM-L6-v2) and FAISS, achieving sub-100ms nearest-neighbor retrieval over 18,000+ documents from the 20 Newsgroups dataset.
- Implemented Gaussian Mixture Model (GMM)-based fuzzy clustering to assign probabilistic multi-topic memberships to documents, improving retrieval relevance for ambiguous queries.
- Designed a semantic cache layer to detect similar queries and reuse results, reducing redundant API calls and cutting average response time by ~40%.
- Exposed REST API endpoints via FastAPI and containerized the full application using Docker, enabling one-command portable deployment.

Digital Library Cloud Platform | React 18, Vite, Supabase, PostgreSQL, Docker, GitHub Actions | [Live Demo](#)

- **Architected & Migrated Full-Stack Platform:** Migrated legacy static application into a modern React 18 & Vite SPA with React Router, implementing modular component architecture, Context API state management, and real-time catalog search.
- **Database & Concurrency Engineering:** Built a relational backend on Supabase (PostgreSQL) with partial unique indexes (unique_active_book_borrowing) to eliminate double-borrow race conditions across concurrent users.
- **Authentication & Role-Based Access Control (RBAC):** Integrated Supabase GoTrue Auth and database Row-Level Security (RLS), supporting dual-identifier login (Username/Email via PL/pgSQL RPC) and route-level protection for members and admins.
- **Testing & Automated CI/CD:** Established zero-dependency automated unit/service test suites (node:test), multi-stage Docker containerization (Alpine/Nginx), and GitHub Actions CI pipelines with edge deployment on Vercel.

TECHNICAL SKILLS

Languages: Python, C, C++, Java, R, JavaScript

Web & Backend: HTML, CSS, React, Supabase, PostgreSQL, FastAPI, REST APIs

ML / AI: Scikit-learn, NumPy, Pandas, Sentence Transformers, FAISS, NLP, RAG, LLMs, Generative AI

AI Concepts: Embeddings, Semantic Search, Vector Databases, Clustering

Tools & Platforms: Docker, Git, GitHub, Vercel, MySQL, Oracle Cloud Infrastructure

Software Engineering: Agile/Scrum, Software Development, UML

CERTIFICATIONS

Hashgraph Developer Course — The Hashgraph Association | Mar 2026

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional — Oracle University | Oct 2025